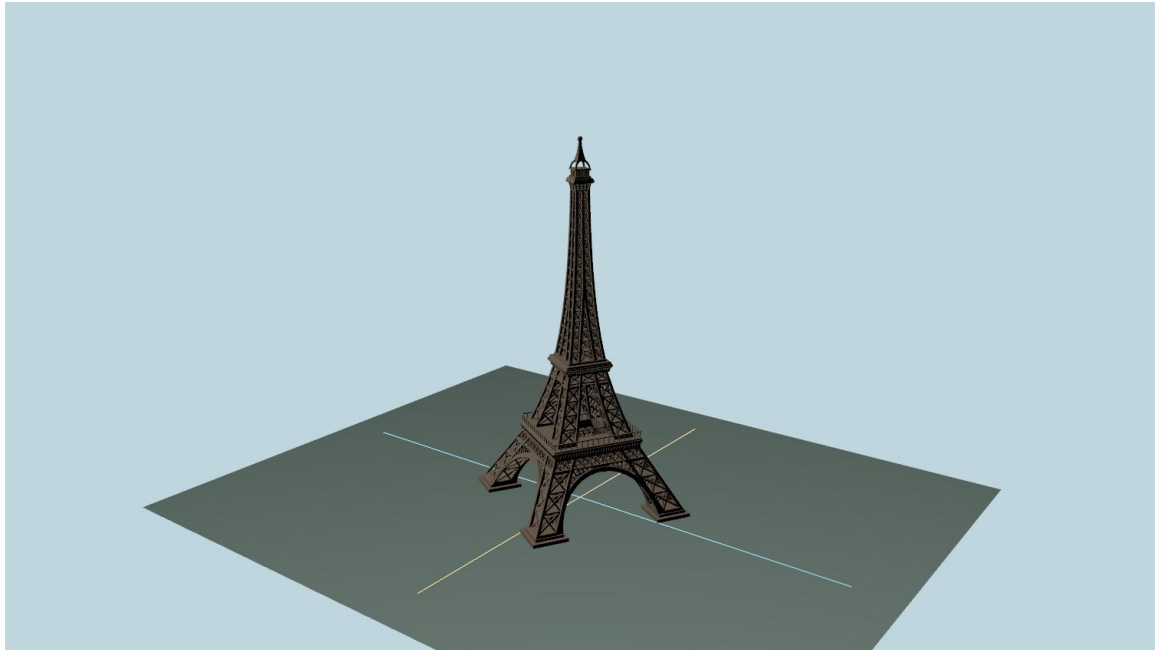
Geospatial USD: proposal quality

A proposal that can drive a fresh implementation

The tested scene semantics were derivable without adding source fields.

Implementation errors were exposed and corrected against the proposal.



C++ placement, ordinary USD bake, native Storm

Bounded evidence of proposal quality. Profiles integration and broader conformance coverage remain explicit.

Inputs describe meaning; readers compute results

Local points retain their ordinary USD length interpretation. Geospatial inputs tell a reader where and how that model is placed.

USD input	Meaning
crs:wkt	Complete CRS description: datum, axes, units, projection and height reference.
crs:position	Absolute origin in its CRS. Geographic 3D uses longitude, latitude and height.
crs:orientation	Physical heading, pitch and roll against local geodetic east, north and up.
Ordinary xformOps	Intentional scale, pivots and adjustments under the specified ordering.
Native dataset association	Asset, format, field and coordinate domain; native values stay paired with locations.

No computed geospatial results become source USD properties. There is no crs:scale or coordinate-epoch field.

Placement and ordinary USD adjustment order

The distinction is between local detail transforms and the anchor's adjustment after intrinsic geospatial placement.

Phase	What happens	Driven by
1. Source evaluation	Evaluate composed origin and physical attitude at the requested time.	position, orientation, Core time rules
2. Local detail	Apply descendant detail transforms, stage units and up-axis mapping.	ordinary local transforms
3. Geospatial placement	Place and orient the detail; honor the full CRS and height operation.	WKT + source position + attitude
4. Working adjustment	Apply the anchor's ordinary transforms in its fixed working chart.	anchor xformOps, including pivots
5. Requested result	Express coordinates in the requested CRS or ordinary scene chart.	output request + numeric origin

A direct child binding is another absolute anchor. Ancestor ordinary transforms do not add its placement again.

WKT string normalization and CRS equivalence

A consistent stored spelling enables exact token comparison. It does not turn every equivalent CRS description into identical text.

Question	Proposed behavior and executed control
Does spelling match?	Normalize spaces, preferred keywords and exact decimal spelling. Preserve quoted content.
Does meaning survive?	Retain axes, units and authored operations. An analytic static-shift control preserves the WKT transformation.
Are different CRSs equivalent?	Resolve semantic equivalence separately. A different normalized token alone does not establish a different Earth reference.

Writers store normalized tokens. Supported non-normal reader input warns and leaves the source layer unchanged.

Composition and repeated georeferenced assets

Sharing a CRS definition and positioning an occurrence have distinct meanings. The controls distinguish those meanings explicitly.

Case	Observed behavior
Shared CRS library	Reference and inherit composition give the same placement in both readers.
Directly bound child	Its absolute placement ignores a parent’s 1,000 m ordinary adjustment.
Repeated georeferenced prototype	A 10 m east instance offset applies once. Storage ancestors do not move the occurrence.
Missing occurrence frame	An identity occurrence keeps placement. A nonidentity adjustment without its required frame fails.

Instance results retain ID/index and prototype context. General inverse and instance-normal coverage remain explicit limits.

USD Profiles: dependency before payload loading

Profiles uses ClaimsAPI metadata on the entrypoint to declare required processing before a consumer loads content.

Author	Publisher	Consumer
Bind a model or native measurement.	Maintain a conservative summary for all retained content and variants.	Read the qualified usage declaration; load no payload to discover it.
Schemas imply the proposed capability.	Publish the summary on the entrypoint, outside payloads.	Treat missing completeness as unknown and retain the dependency.

Proposed identities

usd.geospatial.crsResolution.v1 / usd.geospatial.publication.v1

Publisher coverage is a checkable claim. Discovery does not prove the claim true.

Profiles declarations: executed controls

Both freshly written readers discover the same declaration states with zero loaded payloads.

Published state	Python / C++ discovery	Publisher audit
Retained bound content	required / required	Valid hard dependency
Qualified empty usage	known-none / known-none	Valid when no retained use needs resolution
Unqualified absence	unknown / unknown	Conservative dependency retained
Stale empty summary	known-none claim / same claim	Rejected: a retained variant is CRS-bound

Clarify integration: generic compatibility-query status is not dependency discovery. Supply the final registration graph.

Provider coordinates supply independent expectations

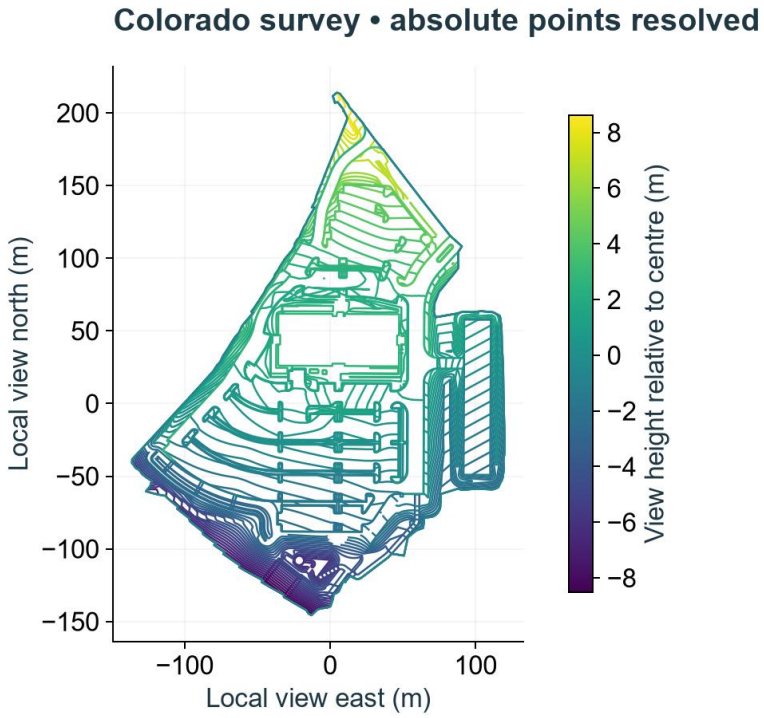
The expected answers come from matched provider point IDs. Reader agreement is reported separately from provider error and operation accuracy.

Provider source	Points	Largest provider error	Python–C++ difference
France 01	59	0.082 mm	0 m in recorded output
France 02	59	0.109 mm	0 m in recorded output
France 04	59	0.100 mm	0 m in recorded output
Colorado 02	5	0.021 mm	0 m in recorded output
Colorado 03	5	0.022 mm	0 m in recorded output

187 comparisons; all within a stated 2 mm tolerance. Rounded controls do not certify survey accuracy.

A site survey keeps absolute-coordinate meaning

Projection and site calibration stay in the complete WKT. Survey vertices are not reinterpreted as local model offsets.



14,359 vertices
710 line parts

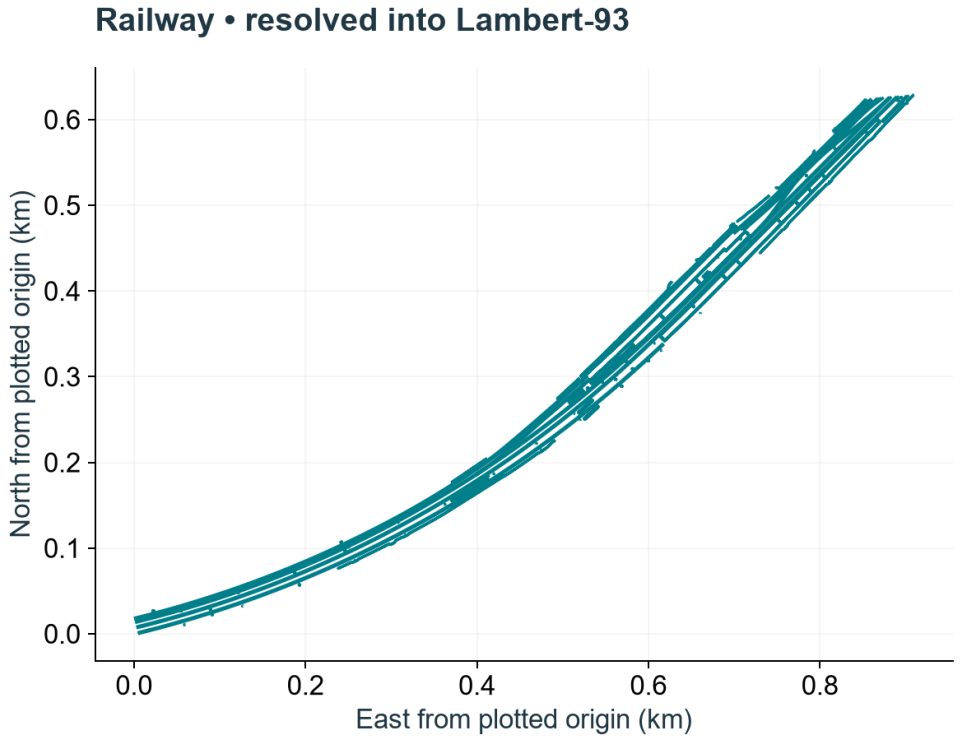
Explicit north/east/height and US survey feet.

The two readers agree on every recorded point.

LandXML is a producer import demonstration, not a fourth schema data format. The view is a plotting convention.

The railway has a useful horizontal result

An unconfirmed height reference does not justify inventing an altitude. The horizontal workflow can still resolve its location.



15,822 coordinates

Explicit 2D derivative.

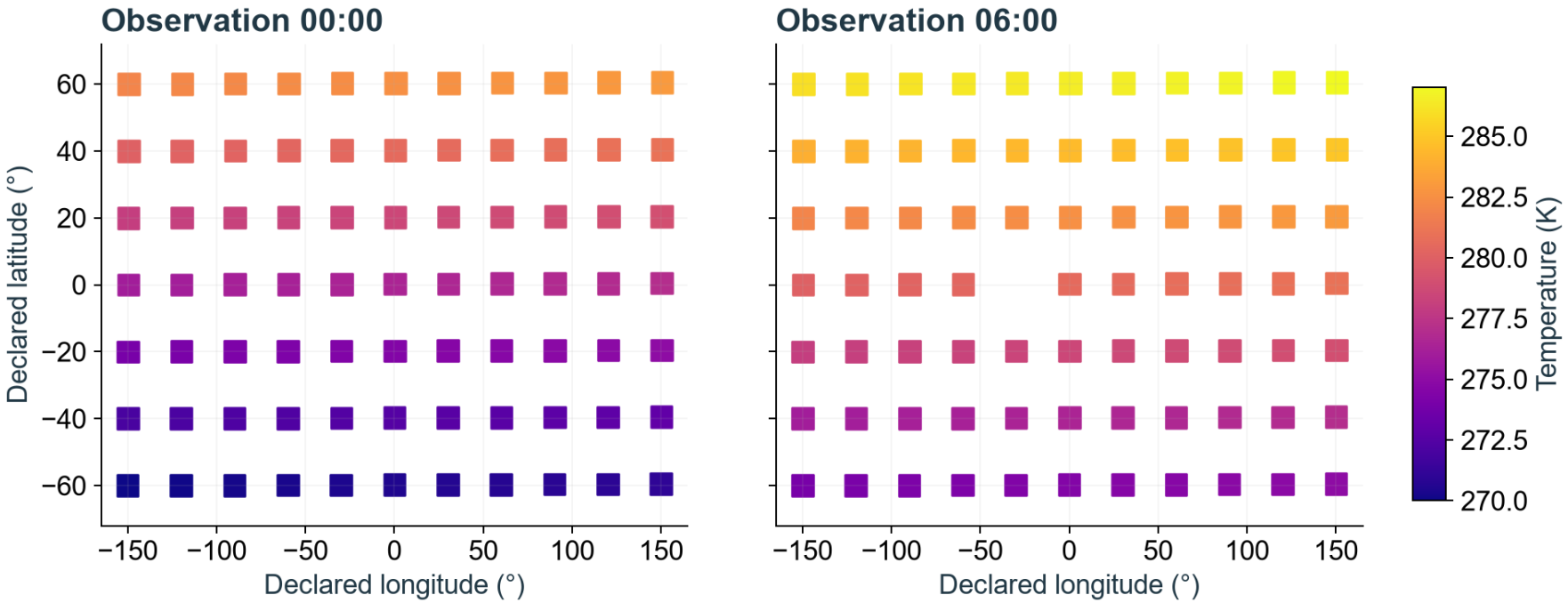
Original third components and file bytes are preserved.

3D meaning remains unclaimed.

The readers agree on the returned horizontal coordinates. No zero-height substitute is used.

CF measurements preserve pairing across space and time

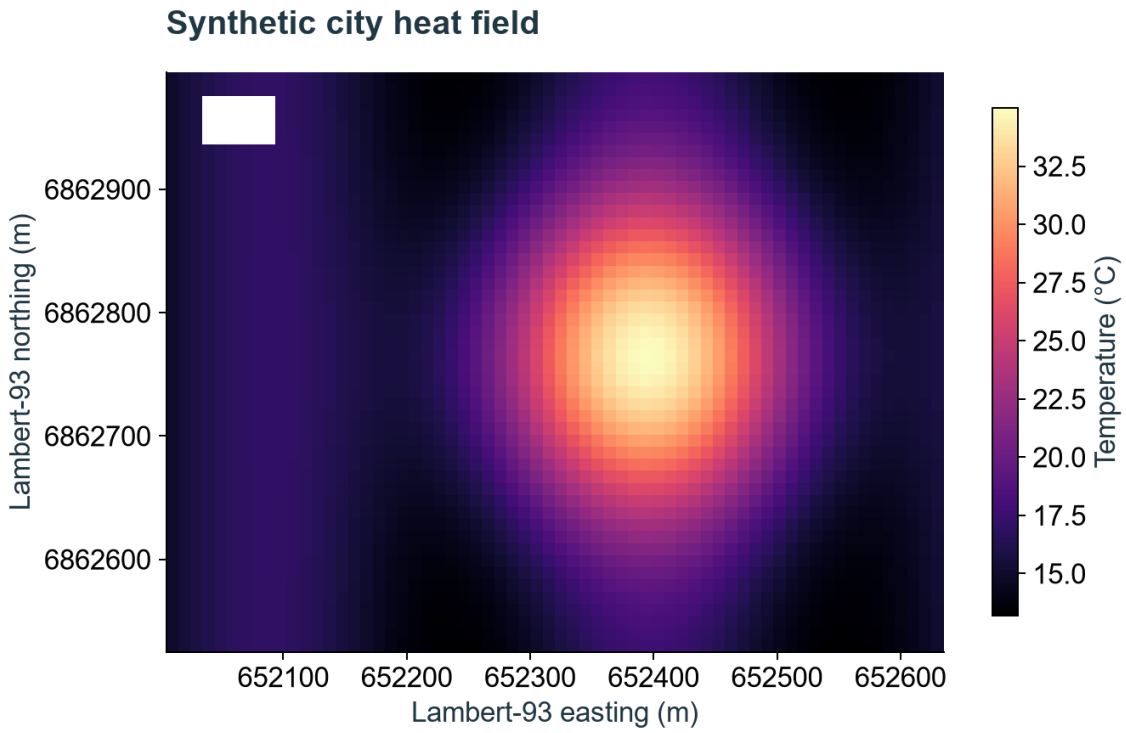
Named dimensions, explicit coordinate roles and observation times determine which value belongs at each location.



154 samples; one masked. The original climate file lacks required metadata and is rejected. This CF example is synthetic.

GeoTIFF analytics retain the native sample domain

USD names the asset, band and IFD. Location resolution preserves the cell’s value and mask.



3,072 cells
24 masked

3,048 valid sample locations agree.

Pixel-centre mapping and missing-height failure have separate controls.

Synthetic 2D heat field; no terrain height assumed. Analytics presentation, not a scene-render claim.

Two placement readers; explicit integration paths

The integrations demonstrate how resolved data can be consumed. They do not add a third placement implementation.

Path	What actually ran
Python placement	Fresh composed-USD reader with PROJ.
Native C++ placement	Fresh native USD reader; independently written placement math and PROJ calls.
Omniverse writeback	Fresh Python results bridged into a stage and read back: 163,416 Eiffel vertices.
Native Hydra / Storm	Actual rendering of ordinary USD geometry baked from the C++ placement results.

Dense reader difference ≤ 8 nm; float32 write error ≤ 0.011 mm. These are different from geodetic accuracy.

The audit tests the implementation against the proposal

Passing images were not enough. The reverse audit checked authored fields, omitted inputs, sampling, native metadata and deferred scope.

Obligation	Distinguishing control
WKT preserves source meaning	OGC-preferred keywords, exact decimals, quoted content and non-normal warnings.
No invented height or date	Short 3D tuples fail; time-dependent operations without coordinate epoch are deferred.
Correct placement roles and order	Invalid binding roles fail; nested anchors, resets, clips and independent instances execute.
A bake represents only what it claims	Unrequested source samples are removed; selected samples and double origin reread correctly.

77 Python controls and 20 native controls, plus provider, sharing and fresh-export evidence.

Proposal readiness and proof limits

The tested semantics can be reviewed on their own merits. A successful bounded demonstration is not complete implementation conformance.

Disposition	Assessment
Scene meaning exercised	No new data-model or placement-rule gap found in the tested paths.
Profiles integration	Make discovery versus generic compatibility explicit; supply and review the registration graph.
Prototype coverage	Nonlinear derivatives, continuous bounds, general instance normals and full format/variant coverage remain unproven.
Deferred scope	Coordinate epochs remain deferred; no implicit epoch is supplied and the model keeps a future path.

Unsupported requests remain visible failures. Evidence does not authorize silently approximating an untested domain.

Fresh derivation and reproducibility

The proposal was the authority. Original data and neutral libraries were reused; previous geospatial source and results were excluded until the first freeze.

New source

Schemas, placement evaluation, adapters and fixture writers were authored anew.

Preserved inputs

Proposal and original dataset hashes are unchanged. Corrective candidate generations retain provenance.

Shared dependencies and scope

USD supplies composition; PROJ supplies geodetic operations. One agent’s fresh derivation is not a blinded independent-author study.

Pinned proposal: e519199. README, deck, source-to-clause trace and executable receipts describe the same bounded evidence.